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Paula Longhi

m.longhi@qmul.ac.uk

William Harvey Research Institute, Barts and the London, Queen Mary University of London, London, UK https://orcid.org/0000-0003-1854-4594

Balraj Sandhar

William Harvey Research Institute, Barts and the London, Queen Mary University of London, London, UK https://orcid.org/0000-0001-7569-0163

Vishal Vyas

Queen Mary University of London

Daniel Harding

William Harvey Research Institute, Barts and the London, Queen Mary University of London, London, UK

Roberta Ragazzini

The Francis Crick Institute https://orcid.org/0000-0003-2186-293X

Paola Bonfanti

Francis Crick Institute https://orcid.org/0000-0001-9655-3766

Federica Marelli-Berg

Queen Mary University of London https://orcid.org/0000-0001-8747-5823

Christopher Bell

Queen Mary Univeristy of London https://orcid.org/0000-0003-4601-1242

Benny Chain

University College London https://orcid.org/0000-0002-7417-3970

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Heterogeneity of thymic output in elderly and its association with sex and smoking.

Balraj Sandhar[1], Vishal Vyas[1, 2], Daniel Harding[1, 2], Roberta Ragazzini [3, 4], Paola Bonfanti [3, 4], Federica M. Marelli-Berg[1], Christopher G. Bell[1], Benny Chain[5], M. Paula Longhi[1].

[1] William Harvey Research Institute, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, London, United Kingdom.

[2] Department of Cardiology, Barts Heart Centre, St Bartholomew's Hospital, London, United Kingdom.

[3] Epithelial Stem Cell Biology & Regenerative Medicine laboratory, The Francis Crick Institute, 1 Midland Road, London NW1 1AT, UK

[4] Institute of Immunity & Transplantation, Division of Infection & Immunity, UCL, Pears Building, Rosslyn Hill, London NW3 2PP, UK

[5] Division of Infection and Immunity, University College London, London, United Kingdom.

Correspondence:

M. Paula Longhi

Address: Heart Centre, William Harvey Research Institute, Barts and The London School of Medicine and Dentistry, Queen Mary University of London, Charterhouse Square, London, EC1M 6BQ

Fax:

Tel: +44 (0) 207 882 6566

Email: m.longhi@qmul.ac.uk

Abstract

Thymic involution is an age-related process that results in reduced T cell production and impaired adaptive immunity. However, it remains uncertain the extent of thymic activity later in life and how this contributes to immunological ageing. Here, we identified thymic tissue within mediastinal adipose locations in aged individuals and confirmed its functional capacity through multi-omics analysis. Percentage of $CD31^+$ $CD4^+$ T cells correlated positively with thymic tissue presence. Thymic output in elderly showed a high degree of heterogeneity was modulated by gender and smoking history. Thymus activity was associated with a diverse TCR repertoire, unique transcriptomic profile, and distinct epigenetic signature. Adipose tissue inflammation was found to be independent of thymic activity but associated with the presence of circulating T cells with an immunosenescent phenotype. Further analysis revealed functional and phenotypic heterogeneity within distinct naïve CD4 T cell subsets, which are influenced by thymic activity.

Introduction

As a primary lymphoid organ, the thymus has a significant role in guiding the differentiation of lymphocyte precursors and facilitating their positive and negative selection processes[1–3]. Therefore, as the fundamental site for T cell development, the thymus establishes adaptive immunity and central tolerance by generating mature immunocompetent T cells. However, over an individual's lifespan, the thymus undergoes involution, a complex process involving structural changes and disorganisation to cortical and medullary compartments accompanied by increased fatty deposition[4,5]. Despite progressive fatty replacement of the thymus with age[6,7], residual thymic tissue can be detected later in life. The degree of thymic activity in older age is less clear with some reports showing minimal function[8,9] while others reporting variable residual thymus tissue after 50 years of age ranging from 16% to 50% [10–13]. The decrease in thymic output during ageing has significant implications for the immune system's functionality[14]. In humans, as thymic activity diminishes, the replenishment of naïve T cell populations in the periphery largely depends on homeostatic proliferation, a process that maintains T cell numbers by cell division[15,16]. This dynamic balance between thymic output and peripheral homeostasis becomes increasingly crucial in the context of immune function, as the gradual decline in naïve T cell numbers is accompanied by oligoclonal expansion of effector/memory T cell subsets[17–20]. Consequently, the overall robustness of adaptive immune responses can be compromised, leading to potential challenges in combating novel pathogens and antigens, which may also concern vaccine response efficacy[21,22].

While age-related pathways have been recognised as a primary driver of thymic involution, recent investigations have unveiled additional factors contributing to accelerated thymic

regression. Studies have correlated markers for metabolic disease, stressors, and sex differences with heightened rates of thymic involution, indicating that the processes influencing thymopoiesis are multifaceted [23–33]. Despite this growing body of research, the role of altered thymic output in older individuals and the underlying factors driving thymus regression later in life, remain poorly understood.

In this study, we sought to characterise thymic function in aged individuals when the rate of thymus involution decreases [9,34]. By analysing distinct T cell populations in blood and correlating these with thymic activity from within mediastinal adipose tissue (AT), we found highly heterogeneous thymus output in individuals over 50 years of age modulated by sex and smoking. Additionally, we assessed markers of thymic function in the blood of patients, correlating these with whole-blood transcriptomic profiles and DNA methylation patterns, finding decreased TCR diversity and increased mortality risk in those individuals with low thymic function.

Results

Functional thymic structures are embedded in the mediastinal adipose tissue.

To effectively evaluate thymic output in aged individuals we first sought to characterise functional thymic tissue obtained from patients (≥ 50 years) undergoing common cardiac surgical procedures, where these tissues could be ethically accessed. Adipose tissues (AT) were collected during surgery from distinct locations within the chest cavity, alongside preoperative fasting blood samples. Specifically, mediastinal AT from superior and inferior regions were acquired for phenotyping (Fig. 1a). These locations were selected as they include sites of the native thymus, involuted thymic remnants, and ectopic thymic deposits[35]. The presence of thymic tissue in AT samples was first assessed by flow cytometry, evidenced by the identification of double positive $CD8^+CD4^+$ (DP) T cells (Fig. 1b). $Thymic^+$ tissues were considered as those having a significant proportion of DP T cells (20-80% of $CD45^+$ cells), with $Thymic^-$ having less than 5% of DP cells from all AT samples tested (Fig. 1b). DP T cells in the superior mediastinal AT could be identified in 48% of patients. From those, a second focus of DP T cells in the lower mediastinal AT was identified in 35% of patients. In 11% of patients, the primary DP focus was present in the lower mediastinal AT. Clinical characteristics of patients are summarised in supplementary Table 1. showing an increased prevalence of females in the $thymic^+$ group. The presence of thymic tissue was confirmed by histological examination as shown in figure 1c. In addition, through different gating strategies (Supplementary Fig. 1a, 1b), the number of mature T cells, conventional dendritic cells (cDC1 and cDC2), plasmacytoid dendritic cells (pDCs) and M2 macrophages were assessed, with

samples with high DP T cells showing increased $CD4^+$ T cells, $CD8^+$ T cells, and cDC2's per gram of tissue (Fig. 1d).

To further validate the thymopoietic properties of thymic$^+$ tissue, bulk RNA-seq and TCR-seq analysis were performed. 813 differentially expressed genes (DEGs) were identified between thymic$^-$ (n = 4) and thymic$^+$ (n = 3) adipose tissue samples (Fig. 1e & Supplementary table 2). Genes associated with development and maintenance of the thymus stromal microenvironment (e.g. *CTSV, FOXN1, PSMB11, TBATA, KRT17, PAX1*) and VDJ recombination (e.g. *RAG1, RAG2, DNTT*) were upregulated in thymic$^+$ samples whereas genes related to adipogenesis (e.g. *ADIPOQ, PPARG, FABP4*) were downregulated. TCRseq analysis further confirmed the functional capacity of thymic$^+$ tissue, mostly through DP T cell populations, in supporting thymocyte differentiation and maturation. This was evidenced by enhanced TRAV and TRAJ usage and combinations (Fig. 1f), and variable CDR3 length usage indicative of immature thymocytes which possess longer CDR3's and $CD3^{hi}$ DP cells with express TCR's with shorter CDR3 sequences following selection (Fig. 1g) [36].

To explore the impact of thymic function on peripheral T cell subsets, the proportion of CD4 and CD8 T cells and their respective phenotypes were assessed in the blood of patients that were initially characterised as either thymic$^+$ or thymic$^-$. Thymic$^+$ patients were shown to have a notable increase in peripheral naïve CD4 T cells compared to thymic$^-$ individuals (Fig. 1h). No significant differences were observed for CD8 mTC proportions between both groups despite the emergence of similar trends (Fig. 1i). Together these findings demonstrate that, despite a decrease in thymic mass and cellularity with age, functional thymic structures could be detected embedded in AT with the capacity to facilitate T cell development.

Thymic function is influenced by sex and smoking status in aged individuals.

Indirect estimates of thymic function are clinically relevant in the context of immune health. Methods such as T cell receptor excision circles (TREC), which are circular excision products formed during TCR rearrangement in the thymus, or quantification of $CD31^+$ naïve $CD4^+$T cells as a surrogate marker of recent thymic emigrants (RTE), have been widely employed to assess thymic function[29,30]. These methods are not without limitations and can be influenced, for example, by cellular expansion within the periphery[37,29,30]. To identify a marker in blood that could correlate with the presence of functional thymic structures in mediastinal AT, we analysed the proportion of $CD31^+$ naïve T cells and TREC values in blood. In addition, to account for changes observed in the $CD4^+$ naïve population (Fig 1h), we tested whether $CD31^+$ $CD4^+$ naïve T cells (referred as RTE%), which are enriched in $TREC^{high}$ T cells [38], could serve as possible putative marker when gated as a proportion of total $CD4^+$ T cells. All three indirect

peripheral measurements of thymic function associated well with the presence of thymic structures, with RTE% having a stronger p value (Fig 2a-b & Supplementary Fig 2a-e). However, only the proportion of $CD31^+$ from total $CD4^+$ T cells, correlated well with TREC values and showed stronger association with DP cells (Supplementary Fig 2f-i). With the validation of RTE% as an easy blood-based marker for thymic output, we expanded our study size (n = 110) with previously processed PBMCs from aged patients. Clinical characteristics are shown in Supplementary table 3. Through linear regression analysis (Fig. 2c), thymic output was shown to correlate positively with naïve T cells and negatively with central and effector memory T cells, consistent with our previous findings with $CD4^+$ T cells (Fig. 1h). Interestingly, no significant correlations were observed between RTE% in blood and effector memory RA^+ T cells (TEMRA), which are reported to have a senescence-associated phenotype[17,39]. The regression of thymus size is not linear but phasic with decreased decline later in life[40,41]. We next evaluated if our cohort followed a similar pattern. Linear regression analysis produced no association between age and RTE % in patients aged over 50 while showing a high degree of heterogeneity in thymic output (Fig. 2d). Under the age of 50, when thymic activity declines at a faster rate, there was a significant negative correlation between RTE % in blood and chronological age ($p = 0.0066$ $R^2 = 0.2428$) supporting previous observations [8,9].

The high heterogeneity observed among older individuals, prompted us to investigate the factors influencing thymic output. Patient characteristics were compared via multiple regression analysis to RTE and naïve T cell %. Variables such as BMI, metabolic status, systemic inflammation, and cardiovascular disease such as ischemia, had no association with RTE % (Supplementary table 4). Interestingly, gender and smoking history were significantly associated with RTE % in contrast to naïve T cell % which fell above the cutoff threshold (Fig. 2f). Levels of recent thymic emigrants were significantly increased in females and for non-smokers (Fig. 2e & 2d). No statistical significance was observed between sex and smoking status in individuals under 50 years of age (Supplementary Fig 2j). These data indicate that sex and smoking can influence the rate of thymic atrophy in aged individuals.

Thymic function is important for a broad T-cell repertoire diversity.

The thymus is critical for the normal development of the immune system in early life. Thymectomy during childhood results in reduced T cell counts, increased oligoclonal memory and declined cellular immunity[42,43]. The health benefits to adults are less clear considering the thymus naturally regresses with age. To investigate the immunological consequences of thymic output in aged patients, we employed a multi-omics approach to further characterise, at the molecular level, the biological changes associated with thymus output (Fig. 3a). An initial

transcriptomic profile of whole blood samples from patients (n = 94) was analysed using microarray analysis to characterise the expression of 19,655 gene probes. Univariate association analysis with RTE % was performed to identify a transcriptomic signature synonymous with recent thymic emigrants. Genes such as *CCR7, PCED1B, TRABD2A, and MMP28* correlated positively to thymic output (Fig. 3b & supplementary table 5). Interestingly, most of these genes have been found to negatively correlate with age[44]. In support of our previous analysis, genes associated with sex, particularly those linked to the Y chromosome, were associated negatively with RTE % (e.g., *UTY, KDM5D, DDX3Y, RPS4Y1*) (Fig. 3b).

Pathway analysis was performed with patients grouped as RTE-high (RTE% ≥20; n = 22) and RTE-low (RTE% ≤10; n = 28) based on the proportion of RTE in their blood. Gene sets associated with a naïve T cell phenotype were enriched in RTE-high patients whereas those related to exhaustion and effector memory T cells were enriched in RTE-low (significant enrichment at FDR < 25%) (Fig. 3c, 3d, supplementary Fig. 3a-c, and supplementary tables 6-8).

To expand our understanding into the immunological impact of thymic activity in aged individuals, we conducted TCR alpha and beta chain sequencing to evaluate T cell receptor repertoire diversity. Shannon diversity index (H) calculations revealed significant correlations between RTE %, but not age, in blood for both alpha ($p = 0.005$ $R^2 = 0.295$) and beta ($p = 0.001$ $R^2 = 0.3827$) chains (Fig. 3e, 3f). Further analysis showed significant positive correlations with RTE % and alpha CDR3's matched to known antigens (McPAS-TCR database) associated with pathogens ($p = 0.0483$ $R^2 = 0.159$), cancer ($p = 0.0175$ $R^2 = 0.222$), and allergy ($p = 0.0316$ $R^2 = 0.186$), which is indicative of higher de novo diversity within the CDR3 [45,46] (Supplementary Fig. 3d).

To test the relevance of a diverse TCR repertoire in ensuring effective immune responses, we generated RTE prediction models to investigate the relationship between RTE levels and severe COVID-19 disease. Three models were generated using 17, 16 or 15 variables achieving similar outcomes (Fig. 3g-i). Of note, genes employed for the predictive model are not associated as classical markers of inflammation or immunity. With all 3 predictive models, patients with severe or critical COVID-19 have significantly lower RTE levels compared to healthy controls or COVID-19 patients not requiring hospital admission (Fig. 3g-i). Collectively, these data show the relevance of thymic output for the efficient control of viral infections and other pathogens.

$CD31^-$ CD4 T cells exhibit an enhanced activated phenotype compared to $CD31^+$ naïve T cells.

Naïve T cells, in particular CD4 subsets, can maintain their levels through homeostatic proliferation, which has been associated with a reduction in $CD31^+$ naïve $CD4^+$ T cells in the periphery and an increase in the proportion of $CD31^-$ mature naïve T cells (MN T cells) [47–49]. Although $CD31^+$ naïve T cells undergo a certain degree of turnover, they are enriched in RTEs[38]. To further explore the immunological impact of thymus function in older age, we evaluated functional differences between MN T cells and RTE's. Transcriptional analysis of publicly available microarray data from sorted human naïve CD4 T cells[50] showed an upregulation of genes linked to T cell activation (e.g. *IL2Rb, CTLA4*) in $CD31^-$ MN T cells (Fig. 4a). Genes linked with RTE phenotype identified from univariate gene association analysis (*MMP28* and *SLC16A10*) (Fig. 3b) were differentially expressed between both cell groups. Interestingly, genes associated with naïve CD4 T cells in the context of ageing, such as *TGFBR3*[51], were also identified amongst DEGs (FDR < 0.05)[51].

Gene set enrichment analysis revealed significant associations with T cell responses and T cell activation via IL2/STAT5 signalling in $CD31^-$ MN T cells (Fig. 4b). With gene sets from GO biological processes, $CD31^-$ MN T cells had a noticeable enrichment for pathways linked to apoptosis and TGFβ production (Fig. 4c), which may be indicative of mechanisms associated with T cell homeostasis within the periphery[52]. Interestingly, RTEs were significantly enriched for oxidative phosphorylation, which could be an indication that these subsets retain a more quiescent T cell state in circulation[53,54].

We next performed functional activation assays using sorted RTE's and $CD31^-$ MN T cells from humans. Naïve T cells were co-cultured with antigen presenting cells (APCs) in the presence of a synthetic superantigen (CytoStim™), which allows for a more physiological activation compared to CD3/CD28 stimulation, and later phenotyped by flow cytometry. After 5 days of culture *in vitro*, $CD31^-$ MN T cells had increased cell surface programmed cell death protein 1 (PD-1) expression which was accompanied by greater proliferation and production of IFNγ compared to RTEs (Fig. 4d-f). Notably, when cultured with IL-7, both subsets demonstrated increased proliferation and cytokine production (Fig. 4e & 4f). Collectively, these findings suggest that MN T cells, despite sharing similar naïve surface markers with RTEs, exhibit distinct functional characteristics, including heightened responsiveness to stimulation and enhanced cytokine production. This implies that while both subsets contribute to the peripheral naïve CD4 T cell pool, they may serve different roles in immune responses.

Adipose tissue inflammation is independent of thymic output in ageing.

Early thymectomy has been associated with marked immunological alterations reminiscent of an aged immune phenotype characterised by immunosenescence and an inflammatory state[16,39,42,43,55,56]. To explore the relationship between thymus output and tissue inflammation in our aged cohort, we assessed T cell subsets and T-cell mediated inflammatory cytokines in epicardial (EAT) and subcutaneous (SAT) adipose tissue by flow cytometry (Fig. 5a-b). EAT represents the fat constituted between the visceral layer of the pericardium and myocardium and is commonly associated with a variety of cardiovascular/metabolic complications[57], while SAT has been associated with age-induced inflammation and senescence [58,59]. Through linear regression analysis, increasing age was found to associate with the percentage of CD4 tissue resident memory T cells (TRMs) in EAT ($p = 0.0224$). Moreover, proportions of CD4 IFNy$^+$ T cells were linked to increased blood CD4 TEMRA ($p = 0.0154$ $R^2 = 0.1772$) and CD8 TEMRA ($p = 0.0092$ $R^2 = 0.2054$) (Fig. 5c), with TEMRA cells often utilised as a blood-based marker for immunosenescence [16,39,56]. No significant correlations were found between inflammatory T cells in EAT and thymic output in patients. Similar findings were observed in the SAT where CD4 IFNy$^+$ correlated positively with CD4 TEMRA and TEM in blood, with no association being observed between RTE% and T cell inflammation (Supplementary Fig. 4) suggesting that peripheral factors, such as environmental and pathogen exposure, are associated with tissue inflammation.

Altered CMV-related T cell profiles in a proportion of patients post-thymectomy have been reported[42]. As CMV has been shown to promote TEMRA differentiation, we next evaluated the impact of CMV infection in our aged cohort. Consistent with previous reports[60–62], our analysis revealed that CMV-positive patients exhibited elevated circulatory levels of both CD4 and CD8 TEMRA cells (Fig. 5d-g), with CD8 factions accumulating at a greater rate with increasing age (Fig. 5h-j). However, we failed to detect any association with RTEs (Fig. 5d,h). Collectively, these findings indicate that tissue inflammation and T cell senescence are independent of thymic function in old individuals.

Low thymic output is associated with accelerated epigenetic ageing.

Thymus removal in adults increases the risk of all-cause mortality and cancer[63]. Epigenetic DNAm clocks are now widely used to quantify and predict 'biological' age and mortality risk[64]. The DNAm GrimAge (AgeAccelGrim), represents a recent and refined second generation 'phenotypic' clock that, in addition to DNAm estimates of age, incorporates estimators of seven ageing-related plasma proteins and pack-years smoking exposure (DNAmPack)[65], thereby improving the prediction of morbidity and mortality and outperforming the first-generation Hannum and Horvath's DNA clocks [66]. To investigate if natural involution of the thymus in old

age associates with increased risk of mortality, we investigated the correlation between DNAm GrimAge and RTEs. We found a significant negative correlation with age acceleration values from GrimAge clock (Fig. 6a & Supplementary Fig.5). No significant differences were observed between smoking status and AgeAccelGrim (Fig. 6b). The association of GrimAge clock with RTE% was attributed to a combined effect of DNAm surrogates for cystatin C (DNAmCystatinC) (Fig. 6c), plasminogen activator inhibitor (DNAmPAI1) (Fig. 6d) and smoking (DNAmPackYears) (Fig. 6e) which are all components of GrimAge. The DNAm estimate of mitotic age (epiTOC2[67]), indicative of cumulative cell division and cancer risk, was also shown to correlate with thymic activity in our cohort (Fig. 6f). Together, these data support the previous association between adults undergoing thymectomy and mortality risk [68–70].

Transcriptomic and epigenomic studies have shown shifts from naïve to differentiated effector/memory T cell states in smokers[71,72]. Prior studies have identified thousands of DNA methylation (DNAm) changes associated with smoking in peripheral blood, of which 62 have been widely reported across multiple studies[73]. To investigate if certain changes could be associated with thymus function in old individuals, we evaluated DNAm array-derived β values for known smoking associated CpGs against RTE % in patient blood through linear regression analysis. Of the 62 CpGs, 54 are present on the Illumina EPIC (850k) array, from which 8 were significantly associated with thymic output: cg21121843 (*HTT*), cg26703534 & cg25648203 (*AHRR*), cg01731783 & cg22851561 (*C14orf43*), cg12803068 (*MYO1G*), cg06060868 (*SDHA*), and cg20295214 (*AVPR1B*) (Fig. 6g). Whilst smoking-related changes in cell proportions contribute to the DNAm signature observed [73], three of these results, cg25648203 (*AHRR*), cg26703534 (*AHRR*), and cg22851561 (C14orf43) were still significant after adjusting for major blood cell type proportions and additional covariates (Fig. 6g). Changes in DNA methylation and gene expression of *GPR15* have been strongly associated with smoking status [71,74]. Although methylation at CpG site cg19859270 of *GPR15*[57,60] failed to reach significance, gene expression signal for *GPR15* negatively correlated with RTE % ($p = 0.0068$ $R^2 = 0.0803$) (Fig. 6h) and was elevated in smokers (Fig. 6i). Furthermore, GPR15 expression was significantly higher in naïve $CD31^-$ CD4 T cells compared to naïve $CD31^+$ CD4 T cells at the protein and RNA level (Fig. 6j & 6k). Indeed, hypomethylation at the cg19859270 site in smokers, is believed to be the result of changes in the proportion of T cell populations in the peripheral blood[75]. Therefore, our data on GPR15 expression supports a link between smoking and thymic output.

Discussion

With global populations ageing at greater rates and declining immunological proficiency being the subject of numerous clinical studies, substantial efforts have been made towards understanding the factors influencing the immune system during ageing [76,77]. Among these factors, thymic involution, a process that impedes the production of new T cells, is thought to contribute to the process of immune ageing. In this study, we conducted a comprehensive examination of thymic output in individuals aged ≥ 50 years. Our findings reveal the persistence of functional thymic structures in adipose tissues from within the mediastinal cavity. These structures possess the capacity to facilitate T cell development, suggesting a complex model in which thymic structures can contribute to peripheral T cell pools whilst being influenced by factors such as sex and smoking status.

Several efforts have been conducted to monitor thymic output in the periphery. We found that the proportions of RTE defined as $CD31^+$ cells from total $CD4^+$ T cells, correlates well with the presence of thymic structures and sjTREC values. In support of previous studies, we show that greater production of these cells correlates with increased naïve T cells and decreased memory T cells for both CD4 and CD8 subsets in the periphery[5]. Chronological age had no influence on thymopoiesis in patients over the age of 50, which aligns with previous reports showing a deceleration of thymic involution after adulthood[78] but was modulated by sex differences and smoking status. Differences in sex were readily evident at the tissue levels with 89% of females being $Thymic^+$, which is consistent with previous observations [79]. Recent work by Marquez et al. has revealed that sexual differences in human immune responses are more pronounced among older individuals with older males displaying a decline in naïve T cell activity compared to age-matched females [80]. Sex-associated hormones such as androgens are known to negatively regulate thymic stromal populations through pathways which impair the survival and differentiation of thymocytes [81–83]. Furthermore, the increased male-bias in diseases associated with impaired immune responses[84], such as COVID-19, could have underlying connections to thymus activity.

The damaging effects of smoking and its association with various pathophysiological complications have been well characterised over the years. Through increased proinflammatory responses, oxidative stress, and DNA damage, smoking can severely restrict both innate and adaptive immune responses[85–87]. The noticeable decrease in recent thymic emigrants observed in smokers, regardless of gender, shows a potential association between thymus proficiency and smoking status. Whether smoking leads to diminished thymus function directly or indirectly through immunological alterations is unclear[88]. Recent studies have found that smoking influences the expression and methylation of GPR15. In particular, increased

expression of GPR15 on T cells has been link to smoking-induced systemic inflammation [75]. We found that *GPR15* gene expression correlated significantly with proportions of recent thymic emigrants in blood. This could be explained, in part, by high expression of GPR15 on $CD31^-$ compared to $CD31^+$ naïve T cells. The functional role of GPR15 remains elusive but it has been implicated in the migration of lymphocytes to large intestine mucosa to maintain homeostatic conditions during inflammation in mice[89,90]. Moreover, GPR15 knock-out models tend to have dysfunctional glucose tolerance, adipose tissue accumulation, and impaired regulation of inflammatory responses[91]. Additional research is required to further understand the role of GPR15 in T cell mediated responses beyond its role as a valuable blood-based marker for smoking-induced immune alterations.

During early life, the post-natal thymus undergoes involution through a process that is highly conserved amongst vertebrates. The evolutionary significance of this process has been widely stipulated as a loss of thymic activity can lead to the development and progression of various disease states. However, it is only recently that the importance of the thymus in adult life has been reported. In a similar cohort of patients, Kooshesh et al found that removal of the thymus increases all-cause mortality while propagating risk for cancer and autoimmune diseases[14]. The negative association with RTE% and the DNAm GrimAge, a mortality risk estimator, confirmed this previous work. The GrimAge clock was constructed with seven DNA methylation-based markers for plasma proteins and smoking packyears [92], which are related to many age-related conditions. In addition to smoking, we found that RTE% associated with two of the plasma proteins; PAI-1 and cystatin C. PAI-1 plays a central role in several age-related subclinical and clinical conditions and have been linked to lifespan in genetic studies[93]. Plasma cystatin C is related to many age-related traits and associated with unsuccessful ageing [94,95]. Cystatin C has been suggested to regulate dendritic cell function and cathepsin S activity, while PAI-1 was shown to promote PDL1 expression, which could have an impact on immunity[96–98]. In addition, individuals with greater thymic output have lower mitotic-count (epiTOC2[67]), which is associated with increased cancer risk, independent of chronological age. The less diverse TCR repertoire in individuals with low RTE%, could conceivably contribute to the immune escape and development of cancer, while Cystatin C and PAI-1 could have an unforeseen role in promoting cancer risk in these patients [99,100].

Here, we show thymus activity in old age influences naïve/memory peripheral T cell pools and TCR repertoire diversity with no impact on inflammageing and immunosenescence as described in early life thymectomy. Thymic output associates with a higher pool of naïve T cells with an enriched fraction of $CD31^+$ cells, with $CD31^-$ naïve T cells having greater

proliferative capacity, activation-induced PD1 expression, and cytokine production (IFNγ & TNFα) when compared to RTE's. It is worth noting that PD1 has been shown to regulate memory T cell differentiation and to limit the generation of central memory T cells, which will have a negative impact on long-term adaptive memory [101]. In addition, greater alpha/beta chain diversity as a result of preserved thymic output would offer improved pathogen recognition and response to a greater variety of novel antigens outside of the preexisting memory T cell population, as has been shown for SARS-CoV-2 vaccine efficacy [102]. Indeed, our predictive model indicates that patients with severe COVID have lower RTEs. However, due to the nature of the study, a direct causal effect cannot be demonstrated.

As this study includes patients undergoing cardiac surgery, which allowed the isolation of tissue samples in the chest cavity, extrapolation of our finding to healthy aged individuals should be done with caution. However, similar sex differences in thymic output have been observed [79], while smoking has been associated with lower thymic CT score in the general population [103]. Furthermore, our study found no association with systemic inflammation (e.g. CRP) or type of surgery. No differences were observed between ischemic (CABG) or valve replacement patients with no heart failure suggesting that the observed thymic outputs heterogeneity is not restricted to our cohort of patients. In addition, a decrease in the frequency of naïve T cells and TCR repertoire diversity, which are hallmarks of thymic function, has been associated with frailty in advanced age. This suggests a degree of heterogeneity in thymus function during ageing across the general population that will have immunological consequences beyond the established aspects of chronological age [104,105].

Methods

Study population and sample collection

Adult patients, undergoing on-pump open chest coronary artery bypass grafting (CABG) surgery, valve surgery (CSS angina class 0 to 1 and a NYHA class 1 to 2) or combined CABG/valve surgery were recruited from Barts Heart Centre, St Bartholomew's Hospital (London, UK) via the Barts BioResource (research ethics committee reference: 14/EE/0007). Patients were screened and then informed written consent was taken for study participation as per local research procedures and Good Clinical Practice guidance. Exclusion criteria included patients with congenital heart disease, underlying cardiomyopathies or ion channelopathies, primarily undergoing other cardiac surgical procedures (e.g. aortic surgery), off-pump CABG surgery, patients with active endocarditis, myocarditis or pericarditis, those with pre-existing inflammatory diseases such as rheumatoid arthritis, active malignancy,

patients on immunomodulatory or biologic drugs (such as tacrolimus or anti-TNF-α agents), peri-operative rhythm control therapies e.g. use of amiodarone, post-operative haemodynamic shock, uncorrected potassium derangement (K<3.3 or K>5.8) or uncorrected magnesium derangement (Mg<0.5 or Mg>1.5) detected on laboratory blood sample analysis. Fasting venous blood samples were collected pre-operatively in the anaesthetic room. Approximately 0.8–1 g of adipose tissue samples were collected in ice-cold phosphate-buffered saline with 2% fetal bovine serum. SAT was collected immediately following the median sternotomy incision and EAT was obtained following opening up of the pericardial sac.

Blood and adipose tissue sample processing

Fasting blood samples were collected preoperatively to include 6.5 mL of peripheral blood divided into 2.5 mL collected in a PAXgene (PreAnalytiX) tube for RNA isolation and the remaining 4 mL in an EDTA tube (BD). PBMCs were isolated using Ficoll-Paque PLUS (GE Healthcare, now Cytiva) as per manufacturer's instructions. PBMCs were then stained using antibodies for flow cytometry analysis. The gating strategy is depicted in Supplemental Figure 1. Following adipose tissue sample collection, samples were divided into a portion for flow cytometry (~0.1–0.4 g) analysis; a portion for subsequent RNA extraction (~0.1–0.2 g), which was snap-frozen; and a sample fixed in 4% formaldehyde solution (MilliporeSigma) for future immunohistochemical analysis (~0.05–0.2 g). The sample of adipose tissue aliquoted for flow cytometry analysis was first mechanically minced using microscopy scissors and then digested enzymatically using 5668 IU collagenase II (MilliporeSigma) and 55.5 IU DNase (MilliporeSigma) per gram of adipose tissue for 30 minutes. Immune cells present in the stromal vascular fraction were obtained following centrifugation and lysed for red blood cells prior to antibody staining.

Flow cytometry staining and analysis

Immune cells from blood and adipose tissue were isolated as described previously. Immune cells were stained with fixable Aqua Live/Dead cell stain (Invitrogen, USA) diluted 1:1000 and fluorochrome-conjugated antibodies specific for CD197-FITC (Biolegend, 353216, 1:200 dilution), CD19-PerCP-Cy5.5 (Biolegend, 302228, 1:200 dilution), CD45RO-BV421 (Biolegend, 304224, 1:200 dilution), CD335-BV605 (Biolegend, 331926, 1:200 dilution), CD45-BV785 (Biolegend, 304048, 1:200 dilution), CD127-APC (Biolegend, 351342, 1:200 dilution), CD8-AF700 (Biolegend, 300920, 1:500 dilution), CD3-APC/Cy7 (Biolegend, 300318, 1:200 dilution), CD69-PE (Biolegend, 310906, 1:200 dilution), CD4-PE/Cy7 (Biolegend, 357410, 1:200 dilution), PD1-PE-CF594 (Biolegend, 329940, 1:200 dilution), KLRG1-SB702

(eBioscience, 15824202, 1:200 dilution), CD303-FITC (Biolegend, 354208, 1:200 dilution), CD123-PerCP/Cy5.5 (Biolegend, 306016, 1:200 dilution), CD206-BV421 (Biolegend, 321126, 1:200 dilution), CD3-BV605 (Biolegend, 317322, 1:200 dilution), CD19-BV605 (Biolegend, 302244, 1:200 dilution), CD14-APC (Biolegend, 325608, 1:200 dilution), CD16-AF700 (Biolegend, 302026, 1:500 dilution), CD1c-APC/Cy7 (Biolegend, 331520, 1:200 dilution), Clec9A-PE (Biolegend, 353804, 1:200 dilution), CD1a-PE-CF594 (Biolegend, 300132, 1:200 dilution), and CD141-PE/Cy7 (Biolegend, 344110, 1:200 dilution). The samples were stained at 4°C for 18 mins and then washed twice with FACS buffer where the plate was centrifuged at 1900rpm for 3 mins and the supernatant discarded. The samples were then fixed in the BD™ stabilizing fixative buffer (BD, USA) containing 3% paraformaldehyde at 4°C for 30 mins. For intracellular cytokine staining, samples were resuspended in 500 µL Aim V medium with the addition of 1 uL of Cell Activation Cocktail (Biolegend, USA). After 4 hours, incubation, samples were washed and stained for surface markers as detailed above followed by permeabilization and fixation in the permeabilization/fixation buffer (BD, USA) at 4°C for 12 mins. Intracellular cytokine production was evaluated by incubation 15 mins at 4°C with fluorochrome-conjugated antibodies specific for IFNy-APC (Biolegend, 502511, 1:200 dilution), IL17-APC/Cy7 (Biolegend, 512319, 1:200 dilution) and IL22-PE (Biolegend, 366703, 1:200 dilution). Data was acquired on a Cytoflex (Beckman Coulter) and analysed using FlowJo version 10 software.

RNA extraction

Total RNA from patient blood samples were extracted using a PAXgeneTM blood RNA kit (catalogue number: 762164). In brief, 2.5 ml of blood was collected into PAXgeneTM RNA tubes and incubated at room temperature for 2 hours. Blood samples were then centrifuged for 10 minutes at 3000g, and supernatants were discarded by decanting. Residual pellets were stored at -80oC for later processing or total RNA was extracted following the manufacturer's instructions immediately. To extract RNA from adipose tissues, samples were finely minced used sterile scissors treated with RNAlaterTM (Invitrogen). To the homogenised tissues, 1 ml QIAzol lysis reagent (Qiagen) was added and thoroughly mixed through inversion. Samples were left at room temperature for 5 minutes and then 200 µl chloroform was added. Total RNA was extracted using RNeasy Lipid Tissue Mini kits (catalogue number: 74804) following the manufacturer's instructions. Relative yields for blood and tissue RNA were quantified using a NanoDrop spectrophotometer.

RT-PCR and Bulk RNAseq

Reverse transcription to cDNA was performed using High-Capacity RNA-to-cDNA kits (Applied Biosystems, Thermo Fisher Scientific) and stored at –80°C. The relevant primer sequences were purchased from Invitrogen, Thermo Fisher Scientific. Gene expression was performed using SYBR Green Supermix (Bio-Rad), as per manufacturer's instructions, and analyzed using the Light Cycler System (Roche). Relative gene expression values were determined using the ΔΔCT method and normalized to a stable reference housekeeping gene control (GAPDH). Illumina sequencing for bulk RNAseq was carried out at Novogene Bioinformatics Technology Ltd.

TREC quantification.

Thymic function was indirectly by measuring sjTREC values in whole blood as previously described [106], with some modifications. Briefly, sjTREC and TRAC fragments were amplified and purified TREC and TRAC PCR products were cloned into the pCR™2.1 vector using a TA Cloning™ kit (Thermo Fisher Scientific). Cloned fragments were used to create a standard curve. gDNA extracted from 200 ul of whole blood and resuspended in 200 μl for qRT-PCR analysis. 2 μl of sample were initially amplified using sjTREC and TRAC primers for twenty cycles to guarantee an accurate amplification at the RT-PCR [107]. Two microlites of a 1/10 dilution of the first round of PCR amplification was further amplified in a qRT-PCR reaction consisting of 2 μl gDNA, 500 μM primers, and 1x iTaq Universal SYBR™ Green, were analysed using a CFX connect light cycler. The copies of sjTRC and TRAC were calculated from the standard curve. The numbers of TRECs were calculated with the formula mean quantity TRECs/ (mean quantity TRAC/2) x 500= copies/ml.

Gene microarray

100ng of total RNA extracted from whole blood were submitted to the University of Bristol Genomics Facility for whole transcriptome expression analysis on the Clariom S Array Plate using the GeneChip™ WT Plus Reagent Kit whilst following the GeneChip™ WT Plus user guide (Catalog numbers 902280 and 902281). Transcriptome analysis Suite v4.0.2.15 was used to perform normalisation and summarisation of the array pegs.

A linear model for RTE% was generated from microarray data. This variant selection method utilised stepwise regression which was employed to identify 3 models using a selection of genes and clinical parameters. For RTE model prediction, count per million (CPM) expression data from Ahern et al.[108] was transformed via $\log_2$ (CMP +1) followed by standard scaling so featured genes had zero mean and unit variance. Raw data has been deposited in GEO (GSE261587).

Methylation microarray & epigenetic clock models

In brief, genomic DNA was extracted from whole blood using the QIAamp DNA blood mini kit (Qiagen) (catalogue number: 51104). DNA quality was assessed Agilent TapeStation. 32 DNA samples were then submitted to University of Bristol Genomics Facility for bisulphite conversion (Zymo Kit)[109] and hybridization on Infinium MethylationEPIC (v1) BeadChip Arrays via the Illumina iScan System using the manufacturer's standard protocol. Raw iDAT files were obtained and standard quality control (QC) and preprocessing was performed via the R package meffil[110]. Briefly, all samples passed QC metrics for correct sex prediction, median methylation intensity, control probe outliers (staining, extension, target removal, hybridization, bisulphite conversion), sample detection (probes with detection p-value > 0.01 is > 0.1), and proportion of probes over bead threshold (bead number < 3 is > 0.1). The DNAm smoking association analysis with RTE was performed via meffil[110] with two regression analysis models i) no covariates and ii) covariates including age, sex, major blood cell types DNAm estimates (Monocytes, Bcell, Neutrophils, CD4T, NK cells)[111], in addition to sva (surrogate variables). The 54 Illumina Epic array available results from the Gao et al. 62 strongly replicated smoking-associated CpGs[73] were then extracted to identify those passing nominal significance ($p \leq 0.05$). Then for DNAm clock analysis the minfi R package[112] was used to implement the recommended probe-type normalization (preprocessNoob). A subset of the 30,084 CpGs were extracted from the total set using the datMiniAnnotation3.csv file for advanced analysis using the online Horvath lab DNAm age calculator https://dnamage.genetics.ucla.edu/[113]. Due to differences in the 850k array from earlier 450k/27k arrays, 2,552 CpGs are not included in this list. The methylCIPHER package was also used to calculate additional DNAm clocks plus DNAm trait estimators[114]. These included the Blood-derived Hannum et al.[115]; Horvath Pan Tissue[113]; PhenoAge [116]; GrimAge[65]; Dunedin Pace of Ageing (PoA) - (DunedinPoAm38) [117]; mitotic EpiTOC2[67] DNAm clocks, and the McCartney et al. estimator of smoking[118].

T cell receptor sequencing

TCR alpha and beta sequencing was performed from cDNA extracted from whole blood and employing a quantitative experimental and computational TCR sequencing pipeline as previously described [119]. The pipeline introduces unique molecular identifiers (UMI) attached to individual cDNA TCR molecule allowing correction for PCR and sequencing errors. TCR identification, error correction and CDR3 extraction was performed following a suite of tools available at github.com/innate2adaptive/Decombinator as detailed in Peacock et al.[120].

T cell stimulation assays

In brief, naïve CD4 T cells were negatively isolated from LSR cones obtained from the NHS using the EasySep™ Human Naïve CD4 T cell isolation kit (StemCell, 19555). Purified cells were stained with AQUA, CD31-APC (Biolegend, 303115, 1:200 dilution), CD4-PE/Cy7 (Biolegend, 357410, 1:200 dilution), and CD8-AF700 (Biolegend, 300920, 1:500 dilution) surface antibodies at 4°C for 18 mins and then washed twice with FACS buffer. Live $CD4^+CD31^+$, $CD4^+CD31^-$, and $CD8^+$ cells were separated by FACS sorting with a LSRFortessa analyser (BD). Antigen presenting cells (APCs) were positively isolated from matched PBMC samples using Biotin anti-human HLA-DR antibody (Biolegend, 307614, 1:200 dilution) and MojoSort™ Streptavidin Nanobeads (Biolegend, 490016). Sorted naïve T cells and APCs were co-cultured in a ratio of 5:1 in AIM-V media supplemented with 20U/ml IL-2 with or without CytoStim™. For proliferation assays, naïve T cells were labelled with 3µM CFSE (Life Technology, C1157) prior to stimulation. After 5 days of culture at 37°C and 5% Co2, cells were stained for intracellular cytokines, as previously described.

Statistical Analysis

Sample size calculations were based on the correlation of RTE% and naïve $CD4^+$ and $CD8^+$ T cells with naïve $CD8^+$ T given the higher sample size of 36 patients with 90% power (α=0.05, effector size=0.489). Statistical significance was determined for continuous variables using 2-tailed Student's t-test where the data were parametric and the Mann-Whitney U test for nonparametric data. Spearman´s correlation was used to analyse univariate associations between quantitative variables. Logistic regression was utilised to deduce R^2 values between continuous variables. The $\chi 2$ test or Fisher's exact test was utilised for categorical data. Data were analysed on GraphPad Prism version 9 (GraphPad Software LLC). Normality was assessed using the Kolmogorov-Smirnov and Shapiro-Wilk tests. Where parametric data are represented, the mean and standard deviation values are reported, and for nonparametric data, median and interquartile ranges are reported.

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Contributions

The manuscript was initially drafted by B.S. and M.P.L. reviewed and edited the manuscript. B.S and M.P.L. contributed to study design and development of concept. B.S. made substantial contribution in data acquisition and analysis. V.V. and D.H. recruited patients and provided clinical data. F.M.B., P.B. and R.R. contributed with resources and interpretation of data for the work. C.B. and B.C. analysed DNAm and TCRseq data respectively. All authors contributed to critical review and revision of the manuscript.

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Figure legends.

Figure 1. Characterising immune cellular landscape and function of thymic structures in aged patients. a) Schematic illustrating project workflow. **b)** Identification of thymocyte populations via flow cytometry in thymic$^+$ (n = 27) and thymic$^-$ (n = 19) samples. Statistical significance was evaluated by two tailed Mann-Whitney U test. **c)** Thymic structure observed through H&E staining. **d)** Immunophenotyping single positive CD4 and CD8 T cells, conventional dendritic cells (cDC1 & cDC2), plasmacytoid dendritic cells (pDC) and M2 macrophages (M2) in thymic$^+$ and thymic$^-$ samples. Statistical significance was evaluated by two-way ANOVA and Sidak test for multiple comparisons. **e)** Volcano plot depicting the average log-fold gene expression changes and Benjamini-Hochberg-corrected p-values for pairwise comparisons between thymic$^-$ control adipose tissue (n = 4) and thymic$^+$ tissues (n = 3). **f)** Representative circos plots illustrating TRAJ (orange) and TRAV (blue) allele pairings from thymic$^+$ and thymic$^-$ samples (n = 1 per group) and **g)** CDR3 length for TCR alpha (TRA) and beta (TRB) chains (thymic$^+$ n = 4; thymic$^-$ n = 4). **h)** Proportion of CD4 and **i)** CD8 naïve, TCM, TEM, and TEMRA cells in the blood of patients classified as either thymic$^+$ or thymic$^-$. Error bars on graph represent mean ± SD. Statistical significance evaluated by two-way ANOVA with Sidak's multiple comparisons test.

Figure 2. Thymic output correlates to peripheral T cell subpopulation and associates with sex and smoking in older patients. a) Flow cytometry gating strategy to identify CD31$^+$ naïve CD4 T cells. **b)** Proportion of recent thymic emigrants in blood of thymic$^+$ and thymic$^-$ patients (thymic$^+$, n = 27; thymic$^-$, n = 19). Error bars on graph represent mean ± SD; significance evaluated by unpaired two-tailed t test. **c)** Linear regression analysis comparing RTE % in blood to peripheral T cell pools. **d)** Correlative linear regression analysis between chronological age and RTE % in blood for individuals aged < 50 years (young; green) (n = 36) and ≥ 50 years (old; purple) (n = 110). **e)** Correlation matrix comparing clinical characteristics to thymic output (relative RTE%) and naïve T cell % in the blood of patients ≥50 years. Spearman coefficients with Bonferroni correction p value cutoff <0.002. Significant correlation coefficients are shown as either blue (positive correlation) or red (negative correlation) dots. **f)** Graph illustrating RTE % in blood between smokers & non-smokers (left) and males & females (right). Truncated violin plots with quartiles, range and with significance evaluated by two tailed Mann-Whitney U test. **c) d)** R^2 values calculated using Pearson correlation coefficients and significance by two-sided p value analysis.

Figure 3. Preserved thymic activity in old patients promotes TCR repertoire diversity. **a)** Schematic illustrating project workflow for the generation of different omics datasets using pre-operative whole blood samples. **b)** Univariate linear regression analysis of gene expression against thymic output (n = 94). **c)** GSEA analysis between patients grouped as RTE low (RTE % ≤10) (n = 28) and RTE high (RTE % ≥20) (n = 22). **d)** Significantly enriched immune GSEA gene sets between RTE high and RTE low patients. **e)** Shannon diversity index (SDI) values for alpha and **f)** beta TCR chains correlated to chronological age and RTE % (n = 25). **g-i)** Stepwise regression models utilising the expression of selected genes and clinical characteristics to predict RTE % in healthy individuals and patients with mild, severe, and critical COVID-19. **b, e-f)** R^2 values calculated using univariate Pearson correlation coefficients and significance by two-sided p value analysis. **g-i)** Statistical significance evaluated by ANOVA with post-hoc Dunn's test. p*<0.05, p**<0.01, p***<0.001, p****<0.0001.

Figure 4. Evaluating the heterogeneity of naïve CD4 T cells. a) Gene expression of *MMP28, SLC16A10, CTLA4, IL2RB*, and *TGFBR3* from humans $CD31^+$ (RTE) and $CD31^-$ (MN) naïve CD4 T cells (n =20). FDR < 0.05. **b)** GSEA pathway analysis between RTEs and MN T cells using Hallmark and **c)** GO biological process gene sets. Plots depict either most enriched, significant, or relevant gene sets. FDR-q < 0.25*. **d)** Plots showing the expression of CD69 and PD-1 on the surface of $CD31^+$ and $CD31^-$ naïve CD4 T cells following stimulation with APC's and CytoStim™. Graph showing $CD69^+PD\text{-}1^-$ and $CD69^-PD\text{-}1^+$ T cells 5 days after stimulation. Error bars on graph represent mean ± SD. Statistical significance evaluated by paired two-tailed t test. **e)** Proliferation index (PI) calculated via CFSE fluorescence co-cultures with or without IL-7. **f)** Histograms showing IFNy and TNFa upregulated in stimulated $CD31^+$ & $CD31^-$ naïve $CD4^+$ T cells and the proportion of $IFNy^+$ T cells co-cultured with or without IL-7. **e) f)** Error bars on bar graph represent mean ± SD; significance between groups evaluated by two-way ANOVA with Tukey's test for multiple comparisons.

Figure 5. Adipose tissue inflammation in old patients is independent of thymic activity. **a) b)** Schematic showing experimental workflow and gating strategy used to characterise inflammatory T cell populations in epicardial adipose tissue (EAT). **c)** Bubble plot showing linear regression analysis comparing T cell % in EAT to blood T cell populations and clinical characteristics. Significant correlations (p < 0.05) are shown as red (positive correlation) or blue (negative correlation) dots. **d-f)** Proportion of RTE, CD8 TEMRA, and CD4 TEMRA in the blood of CMV^+ (n = 55) and CMV^- (n = 52) patients. **g)** Comparison of chronological age

between CMV^+ and CMV^- patients. Volin plots with quartiles, range and significance evaluated by two-sided Mann-Whitney U test. **h)** Linear regression analysis between chronological age and proportion of RTE, **i)** CD8 TEMRA, and **j)** CD4 TEMRA in patients ≥50 years. **c, h-j)** R^2 values calculated using Pearson correlation coefficients and significance by two-sided p value analysis.

Figure 6. Reduced thymic activity associates with epigenetic age acceleration in old patients. a) GrimAge age acceleration (AgeAccelGrim) against RTE % in blood (n = 32). **b)** AgeAccelGrim values compared between male non-smokers (n = 11) and smokers (n = 10). Truncated violin plots with quartiles, range and significance evaluated by unpaired two-tailed t test. **c)** Correlation between RTE % in blood and DNAm surrogates for Cystatin C, **d)** Plasminogen activator inhibitor-1 (PAI1), **e)** smoking (PackYears) and **f)** mitotic age (epiTOC2). **g)** Significant (orange) and non-significant (grey) smoking associated CpG's correlating with RTE %. Model 1: No covariates; Model 2: age, sex and major blood cell type DNAm estimates (monocytes, B cells, neutrophils, CD4 T cells, NK cells). **h)** Linear regression between RTE % and GPR15 gene expression from whole blood microarray data. **i)** GRP15 gene expression from whole blood between patient subgroups defined by sex and smoking status. Violin plots with quartiles, range and significance evaluated by one-way ANOVA with Tukey's test for multiple comparisons. Only significant values are shown. S, Smoker; NS, Non-smoker. **j)** GPR15 gene expression (n = 20) and **k)** Geometric Mean Fluorescence Intensity (gMFI) from $CD31^+$ and $CD31^-$ naïve CD4 T cells (n = 37). Error bars on graph represent mean ± SD; significance by paired two-tailed t test. **a), c-f, h)** R^2 values by Pearson correlation coefficients and significance by two-sided p value analysis.

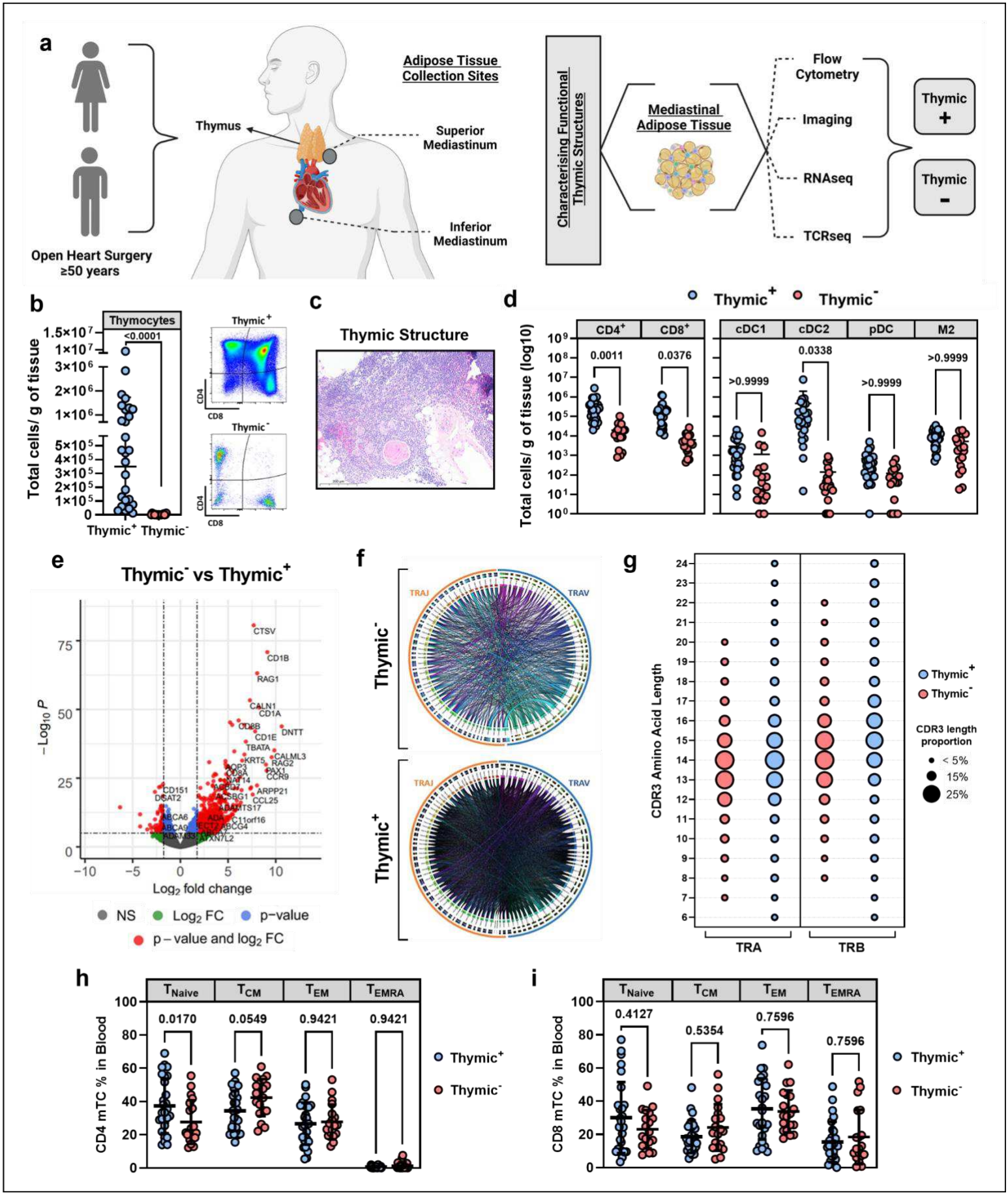

a
Open Heart Surgery
≥50 years
Thymus
Adipose Tissue
Collection Sites
Superior
Mediastinum
Inferior
Mediastinum
Characterising Functional
Thymic Structures
Mediastinal
Adipose Tissue
Flow
Cytometry
Imaging
RNAseq
TCRseq
Thymic
+
Thymic
-
b
Thymocytes
<0.0001
Total cells/ g of tissue
Thymic+ Thymic-
Thymic+
Thymic-
CD4
CD8
c
Thymic Structure
d
Thymic+ Thymic-
CD4+
CD8+
cDC1
cDC2
pDC
M2
0.0011
0.0376
>0.9999
0.0338
>0.9999
>0.9999
Total cells/ g of tissue (log10)
e
Thymic- vs Thymic+
−Log10 P
Log2 fold change
NS
Log2 FC
p-value
p – value and log2 FC
f
Thymic-
Thymic+
TRAJ
TRAV
g
CDR3 Amino Acid Length
Thymic+
Thymic-
CDR3 length
proportion
< 5%
15%
25%
TRA
TRB
h
TNaive
TCM
TEM
TEMRA
0.0170
0.0549
0.9421
0.9421
CD4 mTC % in Blood
Thymic+
Thymic-
i
TNaive
TCM
TEM
TEMRA
0.4127
0.5354
0.7596
0.7596
CD8 mTC % in Blood
Thymic+
Thymic-

Figure 1

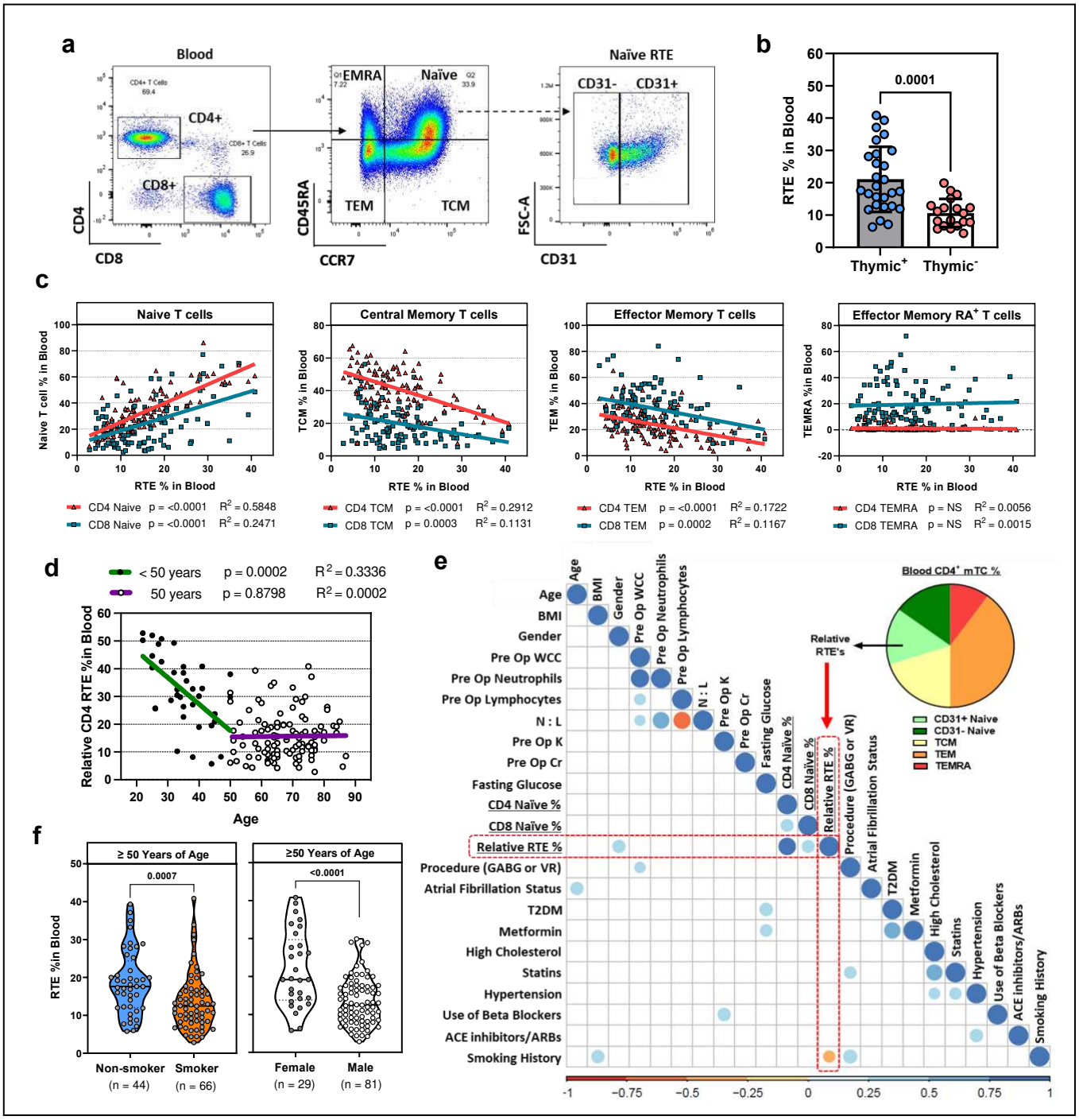

a
Blood
CD4+
CD8+
CD4
CD8
EMRA
Naïve
TEM
TCM
CD45RA
CCR7
Naïve RTE
CD31-
CD31+
FSC-A
CD31
b
RTE % in Blood
0.0001
Thymic+
Thymic-
c
Naive T cells
Central Memory T cells
Effector Memory T cells
Effector Memory RA+ T cells
Naive T cell % in Blood
TCM % in Blood
TEM % in Blood
TEMRA %in Blood
RTE % in Blood
CD4 Naive p = <0.0001 R2 = 0.5848
CD8 Naive p = <0.0001 R2 = 0.2471
CD4 TCM p = <0.0001 R2 = 0.2912
CD8 TCM p = 0.0003 R2 = 0.1131
CD4 TEM p = <0.0001 R2 = 0.1722
CD8 TEM p = 0.0002 R2 = 0.1167
CD4 TEMRA p = NS R2 = 0.0056
CD8 TEMRA p = NS R2 = 0.0015
d
< 50 years p = 0.0002 R2 = 0.3336
50 years p = 0.8798 R2 = 0.0002
Relative CD4 RTE %in Blood
Age
e
Blood CD4+ mTC %
Relative RTE's
CD31+ Naive
CD31- Naive
TCM
TEM
TEMRA
Age
BMI
Gender
Pre Op WCC
Pre Op Neutrophils
Pre Op Lymphocytes
N : L
Pre Op K
Pre Op Cr
Fasting Glucose
CD4 Naïve %
CD8 Naïve %
Relative RTE %
Procedure (GABG or VR)
Atrial Fibrillation Status
T2DM
Metformin
High Cholesterol
Statins
Hypertension
Use of Beta Blockers
ACE inhibitors/ARBs
Smoking History
-1
-0.75
-0.5
-0.25
0
0.25
0.5
0.75
1
f
≥ 50 Years of Age
≥50 Years of Age
RTE %in Blood
0.0007
<0.0001
Non-smoker
(n = 44)
Smoker
(n = 66)
Female
(n = 29)
Male
(n = 81)

Figure 2

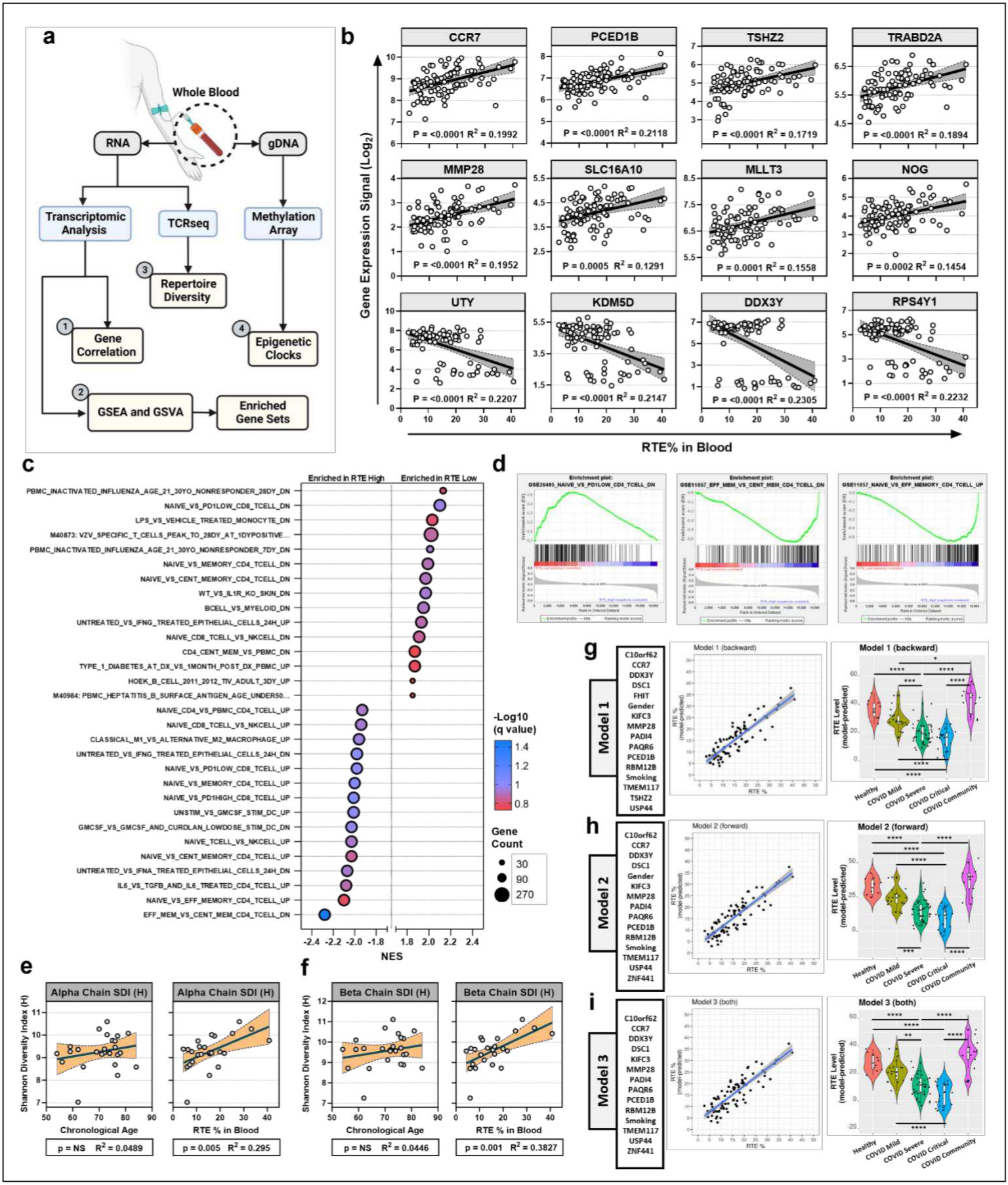
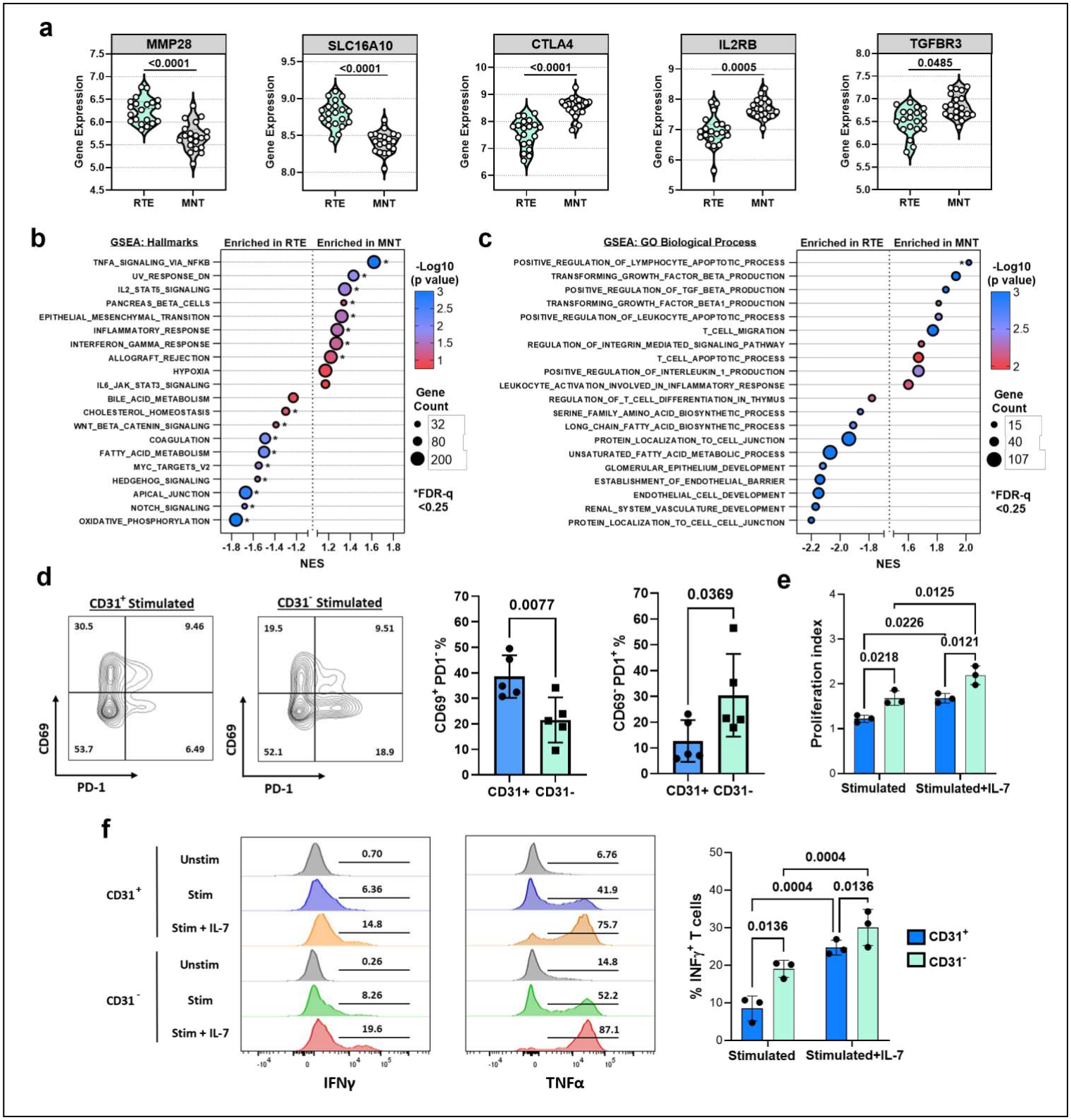

a
Whole Blood
RNA
gDNA
Transcriptomic Analysis
TCRseq
Methylation Array
3
Repertoire Diversity
1
Gene Correlation
4
Epigenetic Clocks
2
GSEA and GSVA
Enriched Gene Sets
b
CCR7
PCED1B
TSHZ2
TRABD2A
MMP28
SLC16A10
MLLT3
NOG
UTY
KDM5D
DDX3Y
RPS4Y1
P = <0.0001 R² = 0.1992
P = <0.0001 R² = 0.2118
P = <0.0001 R² = 0.1719
P = <0.0001 R² = 0.1894
P = <0.0001 R² = 0.1952
P = 0.0005 R² = 0.1291
P = 0.0001 R² = 0.1558
P = 0.0002 R² = 0.1454
P = <0.0001 R² = 0.2207
P = <0.0001 R² = 0.2147
P = <0.0001 R² = 0.2305
P = <0.0001 R² = 0.2232
Gene Expression Signal (Log2)
RTE% in Blood
c
Enriched in RTE High
Enriched in RTE Low
PBMC_INACTIVATED_INFLUENZA_AGE_21_30YO_NONRESPONDER_28DY_DN
NAIVE_VS_PD1LOW_CD8_TCELL_DN
LPS_VS_VEHICLE_TREATED_MONOCYTE_DN
M40873: VZV_SPECIFIC_T_CELLS_PEAK_TO_28DY_AT_1DYPOSITIVE...
PBMC_INACTIVATED_INFLUENZA_AGE_21_30YO_NONRESPONDER_7DY_DN
NAIVE_VS_MEMORY_CD4_TCELL_DN
NAIVE_VS_CENT_MEMORY_CD4_TCELL_DN
WT_VS_IL1R_KO_SKIN_DN
BCELL_VS_MYELOID_DN
UNTREATED_VS_IFNG_TREATED_EPITHELIAL_CELLS_24H_UP
NAIVE_CD8_TCELL_VS_NKCELL_DN
CD4_CENT_MEM_VS_PBMC_DN
TYPE_1_DIABETES_AT_DX_VS_1MONTH_POST_DX_PBMC_UP
HOEK_B_CELL_2011_2012_TIV_ADULT_3DY_UP
M40984: PBMC_HEPTATITIS_B_SURFACE_ANTIGEN_AGE_UNDER50...
NAIVE_CD4_VS_PBMC_CD4_TCELL_UP
NAIVE_CD8_TCELL_VS_NKCELL_UP
CLASSICAL_M1_VS_ALTERNATIVE_M2_MACROPHAGE_UP
UNTREATED_VS_IFNG_TREATED_EPITHELIAL_CELLS_24H_DN
NAIVE_VS_PD1LOW_CD8_TCELL_UP
NAIVE_VS_MEMORY_CD4_TCELL_UP
NAIVE_VS_PD1HIGH_CD8_TCELL_UP
UNSTIM_VS_GMCSF_STIM_DC_UP
GMCSF_VS_GMCSF_AND_CURDLAN_LOWDOSE_STIM_DC_DN
NAIVE_TCELL_VS_NKCELL_UP
NAIVE_VS_CENT_MEMORY_CD4_TCELL_UP
UNTREATED_VS_IFNA_TREATED_EPITHELIAL_CELLS_24H_DN
IL6_VS_TGFB_AND_IL6_TREATED_CD4_TCELL_UP
NAIVE_VS_EFF_MEMORY_CD4_TCELL_UP
EFF_MEM_VS_CENT_MEM_CD4_TCELL_DN
-Log10 (q value)
Gene Count
NES
d
Enrichment plot:
GSE26495_NAIVE_VS_PD1LOW_CD8_TCELL_DN
GSE11057_EFF_MEM_VS_CENT_MEM_CD4_TCELL_DN
GSE11057_NAIVE_VS_EFF_MEMORY_CD4_TCELL_UP
e
Alpha Chain SDI (H)
Shannon Diversity Index (H)
Chronological Age
RTE % in Blood
p = NS R² = 0.0489
p = 0.005 R² = 0.295
f
Beta Chain SDI (H)
p = NS R² = 0.0446
p = 0.001 R² = 0.3827
g
Model 1
C10orf62
CCR7
DDX3Y
DSC1
FHIT
Gender
KIFC3
MMP28
PADI4
PAQR6
PCED1B
RBM12B
Smoking
TMEM117
TSHZ2
USP44
Model 1 (backward)
RTE Level (model-predicted)
RTE %
Healthy
COVID Mild
COVID Severe
COVID Critical
COVID Community
h
Model 2
ZNF441
Model 2 (forward)
i
Model 3
Model 3 (both)

Figure 3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MMP28
<0.0001
SLC16A10
<0.0001
CTLA4
<0.0001
IL2RB
0.0005
TGFBR3
0.0485
Gene Expression
RTE
MNT
b
GSEA: Hallmarks
Enriched in RTE
Enriched in MNT
TNFA_SIGNALING_VIA_NFKB
UV_RESPONSE_DN
IL2_STAT5_SIGNALING
PANCREAS_BETA_CELLS
EPITHELIAL_MESENCHYMAL_TRANSITION
INFLAMMATORY_RESPONSE
INTERFERON_GAMMA_RESPONSE
ALLOGRAFT_REJECTION
HYPOXIA
IL6_JAK_STAT3_SIGNALING
BILE_ACID_METABOLISM
CHOLESTEROL_HOMEOSTASIS
WNT_BETA_CATENIN_SIGNALING
COAGULATION
FATTY_ACID_METABOLISM
MYC_TARGETS_V2
HEDGEHOG_SIGNALING
APICAL_JUNCTION
NOTCH_SIGNALING
OXIDATIVE_PHOSPHORYLATION
-Log10 (p value)
Gene Count
32
80
200
*FDR-q <0.25
NES
c
GSEA: GO Biological Process
POSITIVE_REGULATION_OF_LYMPHOCYTE_APOPTOTIC_PROCESS
TRANSFORMING_GROWTH_FACTOR_BETA_PRODUCTION
POSITIVE_REGULATION_OF_TGF_BETA_PRODUCTION
TRANSFORMING_GROWTH_FACTOR_BETA1_PRODUCTION
POSITIVE_REGULATION_OF_LEUKOCYTE_APOPTOTIC_PROCESS
T_CELL_MIGRATION
REGULATION_OF_INTEGRIN_MEDIATED_SIGNALING_PATHWAY
T_CELL_APOPTOTIC_PROCESS
POSITIVE_REGULATION_OF_INTERLEUKIN_1_PRODUCTION
LEUKOCYTE_ACTIVATION_INVOLVED_IN_INFLAMMATORY_RESPONSE
REGULATION_OF_T_CELL_DIFFERENTIATION_IN_THYMUS
SERINE_FAMILY_AMINO_ACID_BIOSYNTHETIC_PROCESS
LONG_CHAIN_FATTY_ACID_BIOSYNTHETIC_PROCESS
PROTEIN_LOCALIZATION_TO_CELL_JUNCTION
UNSATURATED_FATTY_ACID_METABOLIC_PROCESS
GLOMERULAR_EPITHELIUM_DEVELOPMENT
ESTABLISHMENT_OF_ENDOTHELIAL_BARRIER
ENDOTHELIAL_CELL_DEVELOPMENT
RENAL_SYSTEM_VASCULATURE_DEVELOPMENT
PROTEIN_LOCALIZATION_TO_CELL_CELL_JUNCTION
d
CD31+ Stimulated
CD31- Stimulated
30.5
9.46
53.7
6.49
19.5
9.51
52.1
18.9
CD69
PD-1
CD69+ PD1- %
0.0077
CD69- PD1+ %
0.0369
CD31+
CD31-
e
Proliferation index
0.0125
0.0226
0.0218
0.0121
Stimulated
Stimulated+IL-7
f
CD31+
CD31-
Unstim
Stim
Stim + IL-7
0.70
6.36
14.8
0.26
8.26
19.6
6.76
41.9
75.7
14.8
52.2
87.1
IFNγ
TNFα
% INFγ+ T cells
0.0004
0.0004
0.0136
0.0136
Stimulated
Stimulated+IL-7

Figure 4

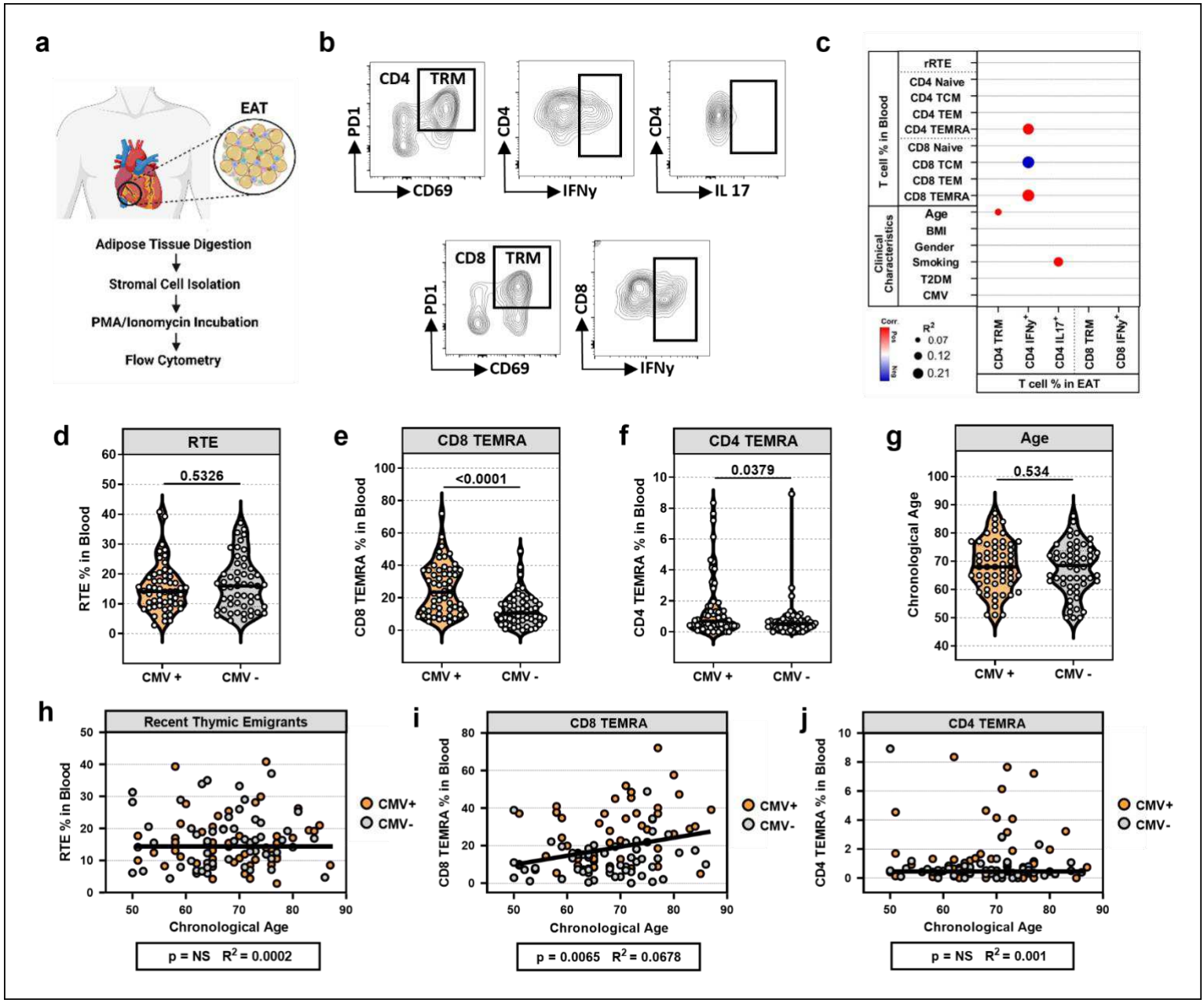
a
EAT
Adipose Tissue Digestion
Stromal Cell Isolation
PMA/Ionomycin Incubation
Flow Cytometry
b
CD4
TRM
PD1
CD69
CD4
IFNy
CD4
IL 17
CD8
TRM
PD1
CD69
CD8
IFNy
c
T cell % in Blood
rRTE
CD4 Naive
CD4 TCM
CD4 TEM
CD4 TEMRA
CD8 Naive
CD8 TCM
CD8 TEM
CD8 TEMRA
Clinical Characteristics
Age
BMI
Gender
Smoking
T2DM
CMV
Corr.
Pos
Neg
R2
0.07
0.12
0.21
CD4 TRM
CD4 IFNy+
CD4 IL17+
CD8 TRM
CD8 IFNy+
T cell % in EAT
d
RTE
0.5326
RTE % in Blood
CMV +
CMV -
e
CD8 TEMRA
<0.0001
CD8 TEMRA % in Blood
CMV +
CMV -
f
CD4 TEMRA
0.0379
CD4 TEMRA % in Blood
CMV +
CMV -
g
Age
0.534
Chronological Age
CMV +
CMV -
h
Recent Thymic Emigrants
RTE % in Blood
Chronological Age
CMV+
CMV-
p = NS R2 = 0.0002
i
CD8 TEMRA
CD8 TEMRA % in Blood
Chronological Age
CMV+
CMV-
p = 0.0065 R2 = 0.0678
j
CD4 TEMRA
CD4 TEMRA % in Blood
Chronological Age
CMV+
CMV-
p = NS R2 = 0.001

Figure 5

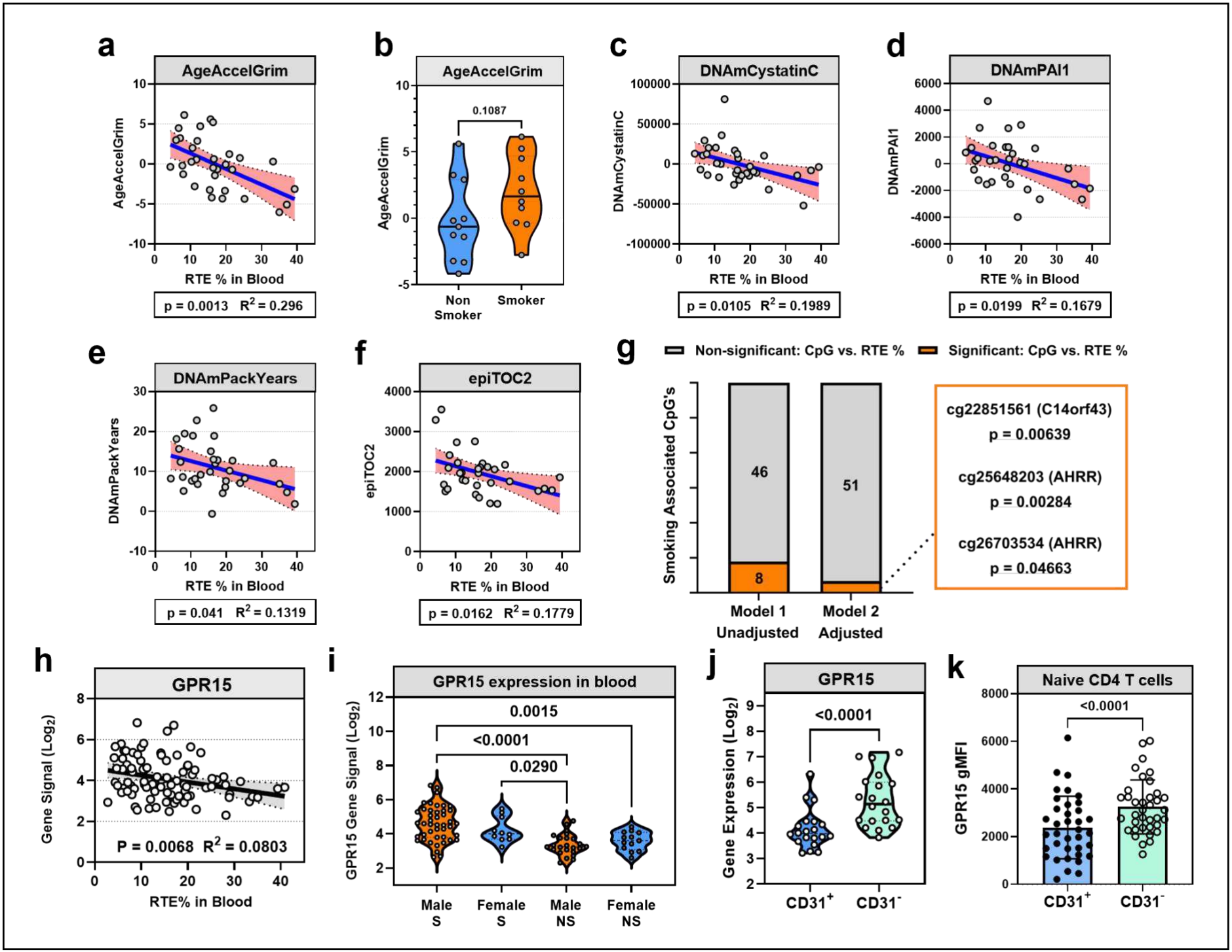
a
AgeAccelGrim
AgeAccelGrim
RTE % in Blood
p = 0.0013 R² = 0.296
b
AgeAccelGrim
0.1087
AgeAccelGrim
Non Smoker
Smoker
c
DNAmCystatinC
DNAmCystatinC
RTE % in Blood
p = 0.0105 R² = 0.1989
d
DNAmPAI1
DNAmPAI1
RTE % in Blood
p = 0.0199 R² = 0.1679
e
DNAmPackYears
DNAmPackYears
RTE % in Blood
p = 0.041 R² = 0.1319
f
epiTOC2
epiTOC2
RTE % in Blood
p = 0.0162 R² = 0.1779
g
Non-significant: CpG vs. RTE %
Significant: CpG vs. RTE %
Smoking Associated CpG's
46
51
8
Model 1 Unadjusted
Model 2 Adjusted
cg22851561 (C14orf43)
p = 0.00639
cg25648203 (AHRR)
p = 0.00284
cg26703534 (AHRR)
p = 0.04663
h
GPR15
Gene Signal (Log2)
P = 0.0068 R² = 0.0803
RTE% in Blood
i
GPR15 expression in blood
GPR15 Gene Signal (Log2)
0.0015
<0.0001
0.0290
Male S
Female S
Male NS
Female NS
j
GPR15
Gene Expression (Log2)
<0.0001
CD31+
CD31-
k
Naive CD4 T cells
GPR15 gMFI
<0.0001
CD31+
CD31-

Figure 6

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Supplfilecombined.pdf